



II Olimpiada Informática de Las Palmas

Contest – January, 26th 2024

Intergalactic Phrase Translator

Translator

Intergalactic Phrase Translator

Problem Statement:

A group of *Earth inhabitants* has arrived on the planet *Thalassa* in the *Zephyria Nebula* galaxy but is having difficulties communicating in the languages of its inhabitants. The *Earthlings* have realized that the two types of inhabitants on this planet, whom they have called *Blues* and *Greens*, use alphabets and languages very similar to Spanish but simplified. The *Blue* language has an alphabet of five letters: “azulo”, and the *Green* language has an alphabet of seven letters “verdosu”. The *Earthlings* have managed to find a translation system based on the substitution of letters in Spanish words with letters from the alphabets of the *Thalassian* languages (Blues and Greens).



Your task is to create a program that translates phrases between Spanish and the languages of the Blue and Green inhabitants by applying specific transformation rules.



II Olimpiada Informática de Las Palmas

Contest – January, 26th 2024

Intergalactic Phrase Translator

Transformation Rules:

- **Language of Blue inhabitants:** each letter of the word in Spanish that occupies the position pos in the Spanish alphabet is transformed into the letter of the Blue alphabet at the position resulting from the remainder of dividing the position in Spanish by five. **Note** the positions in both alphabets start at zero. **For example**, 'a' becomes 'a', 'b' becomes 'z', 'c' becomes 'u', 'e' becomes 'o', 'f' becomes 'a', and so on.

- **Language of Green inhabitants:** each letter of the word in Spanish that occupies the position pos in the Spanish alphabet is transformed into the letter of the Green alphabet at the position resulting from the remainder of dividing the position in Spanish by seven. **Note** the positions in both alphabets start at zero. **For example**, 'a' becomes 'v', 'b' becomes 'e', 'c' becomes 'r', 'e' becomes 'o', 'f' becomes 's', and so on.

Note: It is also possible to translate directly from Spanish to any alien alphabet using extended alien alphabets where the sequence of the original alien alphabet is repeated. In this case, a letter in the Spanish alphabet would be transformed into the letter that is in the same position in the extended alien alphabet. To avoid encoding problems, a simplified Spanish alphabet will be used.

Spanish Alphabet: abcdefghijklmnopqrstuvwxyz

Blue Alphabet: azulo

Green Alphabet: verdosu



II Olimpiada Informática de Las Palmas

Contest – January, 26th 2024

Intergalactic Phrase Translator

Input / Output

Input Format:

The input will consist of a phrase where the first letter indicates whether the phrase should be translated into the language of the Blue inhabitants “A” or into the language of the Green inhabitants “V”. The first letter will always be “A” or “V” in uppercase followed by a blank space; otherwise, the output must be ERROR.

Output Format:

The program will display the translated phrase applying the corresponding transformation rule to each letter, considering Blues or Greens. Spaces are maintained the same, any other character that is not a space and is not found in the alphabet will give an ERROR. The program on the rest of the line must be able to handle both uppercase and lowercase letters.

Input / Output examples:

Example 1

Input:

A abcdefghijklmnopqrstuvwxyz

Output:

azuloazuloazuloazuloazuloa

Example 2

Input:

A hola amigos



II Olimpiada Informática de Las Palmas

Contest – January, 26th 2024

Intergalactic Phrase Translator

Output:

uoza aulzol

Example 3

Input:

R fff

Output:

ERROR